

HEAVEN'S LIGHT IS OUR GUIDE



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF MECHATRONICS ENGINEERING, 21Series

Course No: MTE 3102

Course Title: Control Systems Sessional

Experiment No: 08

Experiment Name: Design of a PID Controller for the systems by using Root Locus Method and Performance Analysis.

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Experiment Name: Design of a PID Controller for the systems by using Root Locus Method and Performance Analysis.

Objectives:

1. To design PID, PD & PI controllers for given systems
2. To analyze and compare the performance of PID, PD & PI controllers through simulation
3. To design a PID controller for the given systems using the root locus method and to analyze and compare its performance characteristics

Theory:

PID Controller: PID control, representing proportional-integral-derivative control, is a feedback mechanism in control system, often referred to as three-term control. By adjusting three parameters—the proportional, integral, and derivative values of a process variable's deviation from its set point—specific control actions are effectively tailored.[1]

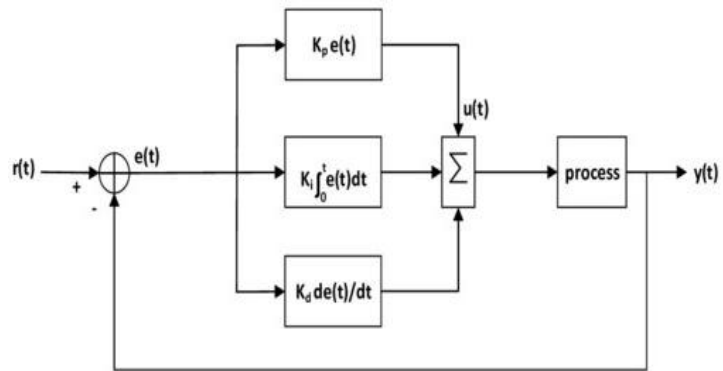


Figure 1: Basic PID controller block diagram. [3]

We need to add a complex zero and a pole to the system.

Proportional (P) response: The proportional gain (K_c) determines the ratio of output response to the error signal. In general, increasing the proportional gain will increase the speed of the control system response.

Integral (I) response: The integral component sums the error term over time. The result is that even a small error term will cause the integral component to increase slowly. The integral response will continually increase over time unless the error is zero, so the effect is to drive the Steady-State error to zero.

Derivative(D) response: The derivative component causes the output to decrease if the process variable is increasing rapidly. The derivative response is proportional to the rate of change of the process variable.[2]

PID Controller Equation:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

In Laplace Domain,

$$G(s) = K_p + \frac{K_i}{s} + K_d s$$

$$PID = \frac{K_p s + K_i + K_d s^2}{s} = \frac{K_d \left(s^2 + \frac{K_p}{K_d} s + \frac{K_i}{K_d} \right)}{s}$$

Control Response	Rise time	Settling time	Overshoot	Steady state error
K_p	decrease	small change	increase	decrease
K_d	small change	decrease	decrease	no change
K_i	decrease	increase	increase	eliminate

Table 1: PID Parameters Summary Table [1]

PI Controller: PI control, a combination in PID systems, doesn't include derivative control. It's a type of feedback control that combines proportional and integral actions. PI control responds faster than integral-only control because it includes proportional action. It helps stabilize the system and brings it back to the set point. [4]

We need to add a zero and a pole at origin to the system.

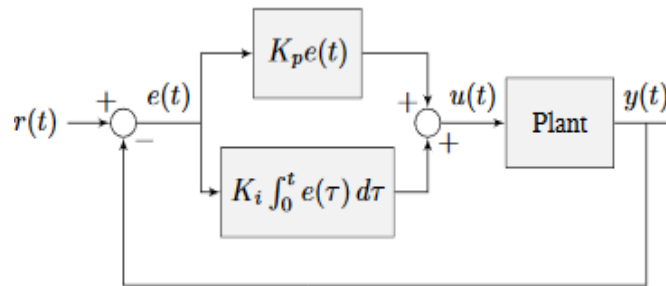


Figure 2: Basic PI controller block diagram. [5]

$$PI = K_p + \frac{K_i}{s} = \frac{K_p s + K_i}{s} = \frac{K_p \left(s + \frac{K_i}{K_p} \right)}{s}$$

PD Controller: PD control, another combination in PID systems, doesn't include integral control. It's a mix of feedforward and feedback control, taking into account current and predicted process conditions. In PD control, the controller output is a combination of the error signal and its derivative. This means it considers both how much the error is right now and how fast it's changing.[4]

We need to add a zero to the system.

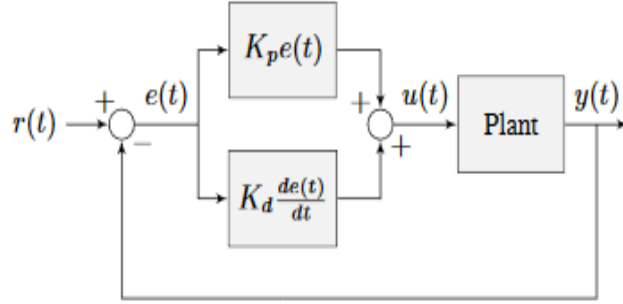


Figure 3: Basic PD controller block diagram. [5]

$$PD = K_p + sK_d = K_d \left(s + \frac{K_p}{K_d} \right)$$

Root Locus: A graphical method used for analyzing the location and movement of poles in the s-plane with the variation in the gain factor of the system is known as Root Locus. This technique is used to check the stability of the closed-loop control system. So, in simple terms, a graph plotted for roots of a characteristic equation by varying the system parameter (generally gain) from 0 to infinity is called Root Locus. Root Locus Analysis was proposed by W R Evans in the year 1948. [6] It is especially useful for designing compensators such as PID controllers in systems like DC motors, where precise performance specifications are required.

Required Apparatus:

MATLAB

System Models:

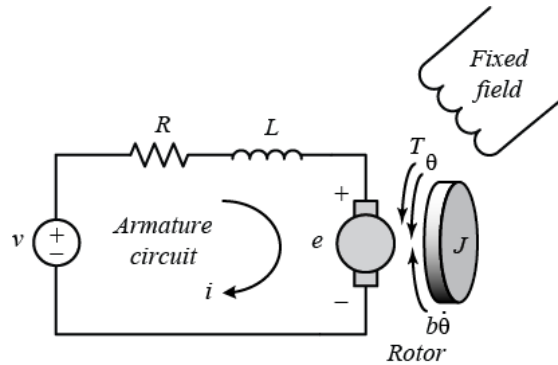


Figure 4: System Representation of DC Motor

Transfer Function:

Motor system transfer function,

$$\frac{\theta(s)}{v(s)} = \frac{K}{\{(Js + b)(Ls + R) + K^2\}}$$

MATLAB Code:

PID

```
clc;
clear all;
%system parameters
J=0.01;
b=0.1;
K=0.01;
R=1;
L=0.5;
s=tf('s');
%Transfer Function Reprastantion
TF=K/((J*s+b)*(L*s+R)+K^2);
%Design of PID Controller
Kp= 11.83;
Ki= 2.183;
Kd= 15.99;
PID=pid(Kp, Ki, Kd);
% Closed-loop System
CL = feedback(TF*PID,1);
%Outputs
figure(1)
rlocus(TF*PID);
figure(2)
step(CL);
disp(stepinfo(CL));
```

PD

```
clc;
clear all;
%system parameters
J=0.01;
b=0.1;
K=0.01;
R=1;
L=0.5;
s=tf('s');
%Transfer Function Reprastantion
TF=K/((J*s+b)*(L*s+R)+K^2);
% %Design of PID Controller
Kp= 9.2638;
Ki= 0;
Kd= 66.17;
PD=pid(Kp, Ki, Kd);
%Closed-loop System
CL = feedback(TF*PD,1);
%Outputs
figure(1)
rlocus(TF*PD);
figure(2)
step(CL);
disp(stepinfo(CL));
```

PI

```
clc;
clear all;
%system parameters
J=0.01;
b=0.1;
K=0.01;
R=1;
L=0.5;
s=tf('s');
%Transfer Function Reprastantion
TF=K/((J*s+b)*(L*s+R)+K^2);
% %Design of PID Controller
Kp= 1.319;
Ki= 5.9738;
Kd= 0;
PI=pid(Kp, Ki, Kd);
%Closed-loop System
CL = feedback(TF*PI,1);
%Outputs
figure(1)
rlocus(TF*PI);
figure(2)
step(CL);
disp(stepinfo(CL));
```

Result:

PID Controller

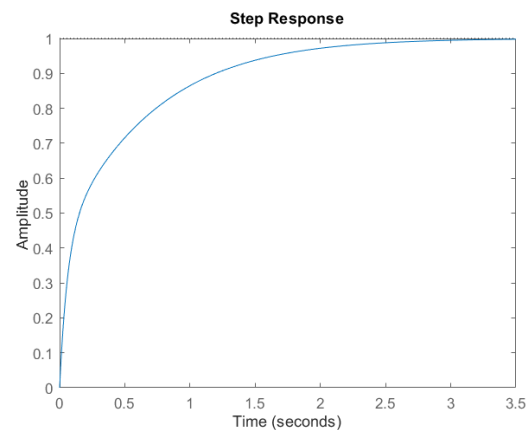
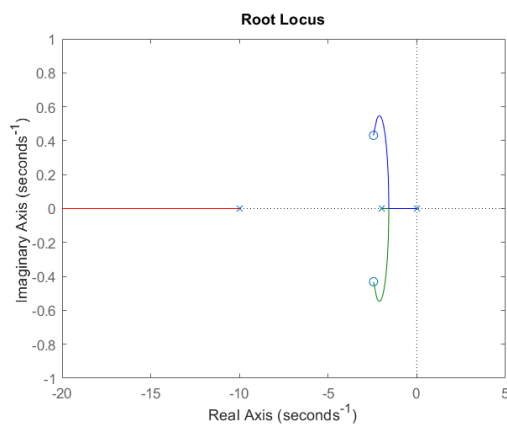


Figure 5: Root Locus Plot & Step response of PID in MATLAB

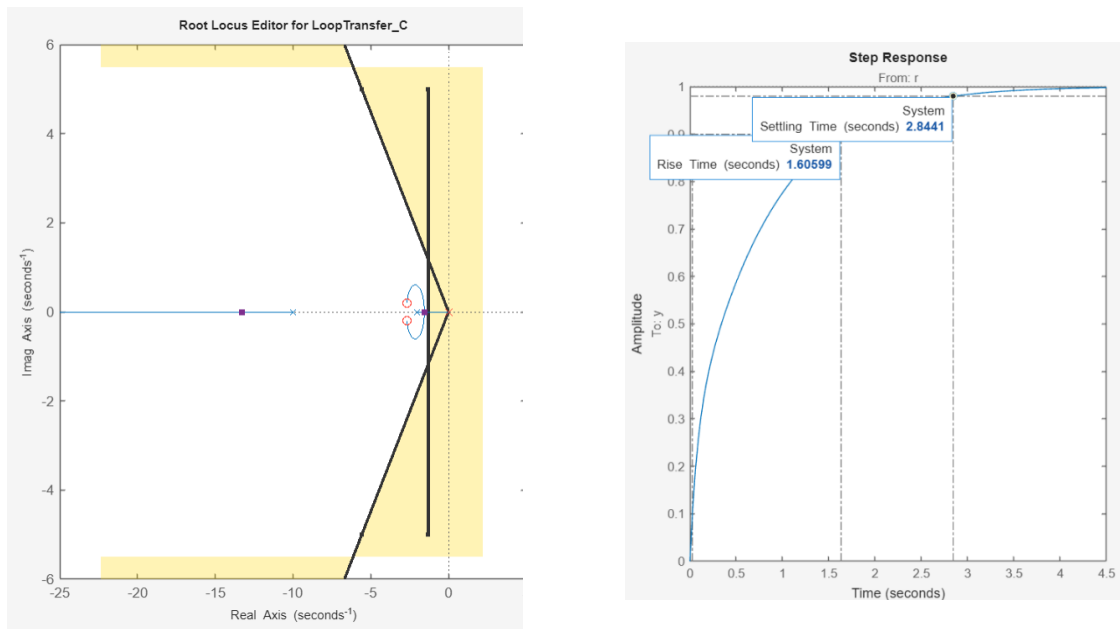


Figure 6: Root Locus Editor and Step response of PID in Control System Designer App

PD Controller

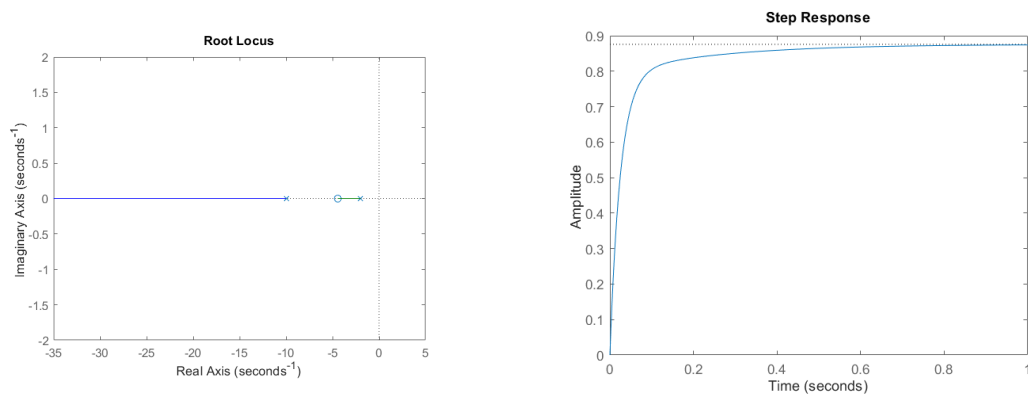


Figure 7: Root Locus Plot & Step response of PD in MATLAB

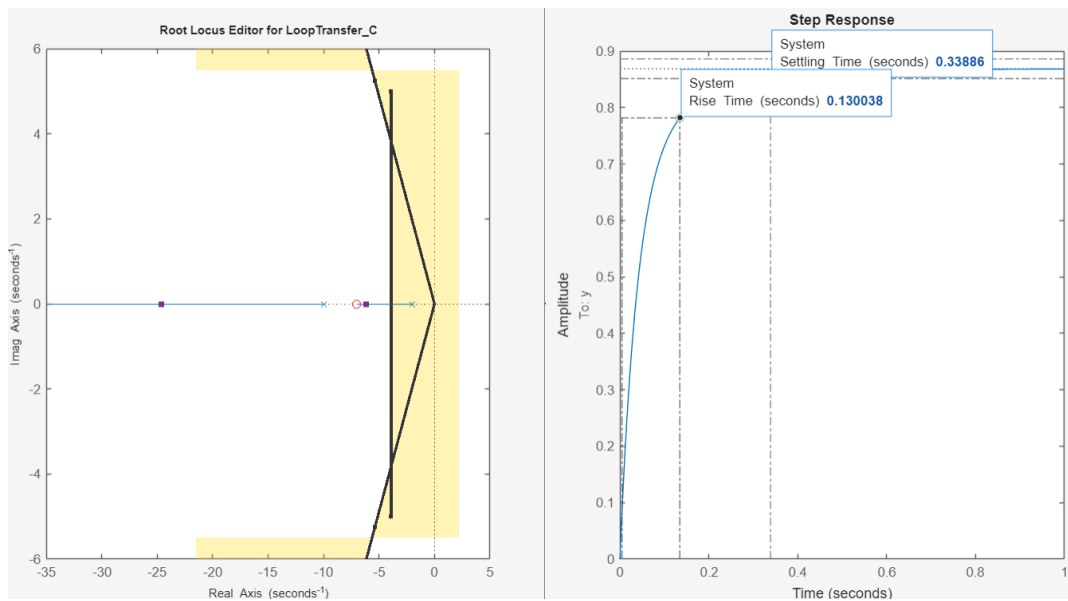


Figure 8: Root Locus Editor and Step response of PD in Control System Designer

PI Controller

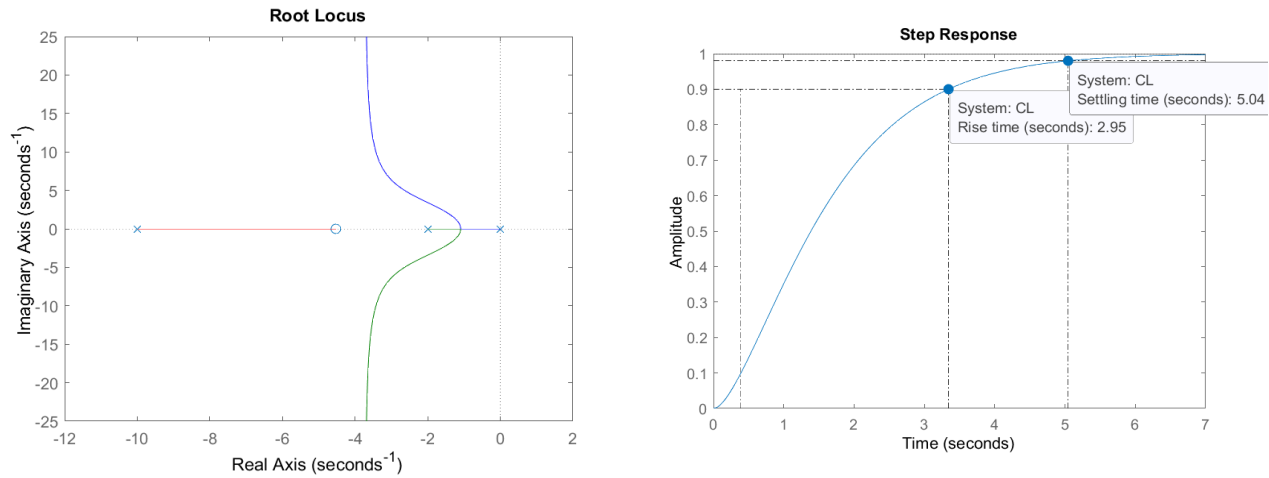


Figure 9: Root Locus Plot & Step response of PD in MATLAB

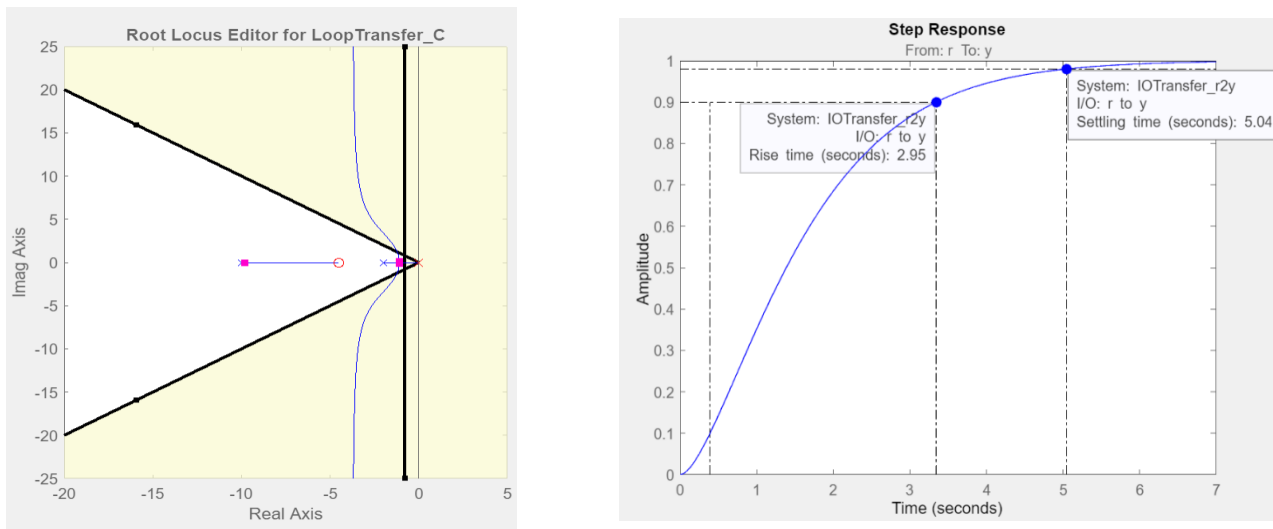


Figure 10: Root Locus Editor and Step response of PD in Control System Designer

Performance Analysis:

Controller	K_p	K_i	K_d	Rise Time	Settling Time	Overshot (%)
PID	11.83	2.183	15.99	1.60599	2.8441	0
PD	9.2638	0	66.17	0.130038	0.33886	0
PI	1.319	5.0973	0	2.9523	5.0441	0

Discussion: The root locus method graphically shows how closed-loop poles shift with varying controller gain, helping assess stability and performance. Performance comparisons reveal that the PD controller has the fastest response but requires high gains. The PI controller eliminates steady-state error with no overshoot but responds more slowly. The PID controller offers a balanced performance with moderate speed and zero overshoot, making it a good compromise between response time and control effort.

Conclusion: The root locus method visually tracks closed-loop pole movement as controller gain changes, aiding in stability analysis. Among PI, PD, and PID controllers, the PD gives the fastest response but demands high gain. The PI eliminates steady-state error with slower response and no overshoot. The PID balances speed and control, making it an effective overall choice.

Reference:

- [1] Electrical4U, "PID control – proportional integral derivative," *Electrical4U*, [Online]. Available: <https://www.electrical4u.com/pid-control/>. Accessed: Apr. 26, 2025
- [2] National Instruments, "PID Theory Explained," *NI*, [Online]. Available: <https://www.ni.com/en/shop/labview/pid-theory-explained.html>. Accessed: Apr. 26, 2025.
- [3] M. Malekabadi, M. Haghparast, and F. Nasiri, "Air Condition's PID Controller Fine-Tuning Using Artificial Neural Networks and Genetic Algorithms," *Computers*, vol. 7, no. 2, p. 32, May 2018, doi: 10.3390/computers7020032.
- [4] Kamboj, A., "Types of Controller | I, D, PD, P, PI, PID Control," *Engineeringa2z*, March 1, 2024. Available: <https://www.engineeringa2z.com/types-of-controller-i-d-pd-p-pi-pid-control/>.
- [5] *Introduction to PID. WPILib Docs. Updated: June 2, 2025. Available: https://docs.wpilib.org/en/stable/docs/software/advanced-controls/introduction/introduction-to-pid.html.*
- [6] Root Locus Construction with Examples, *Electronics Coach*, ca. 4.9 years ago. Available: <https://electronicscoach.com/proportional-integral-controller.html>.